claude-windows / fa4dc4eb-4705-47fd-b044-4f017b586b15

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\fa4dc4eb-4705-47fd-b044-4f017b586b15\subagents\workflows\wf_ac0cbb29-6bf\agent-a0b3e61af60b87433.jsonl`
- SHA-256: `d5156ebd2f9d49eb2308dd93f832c27740f0fdf8d5ea5c7a44647e4bb2533a41`
- Source modified: `2026-06-19T15:53:47+00:00`
- Imported at: `2026-07-05T16:48:22+00:00`
- project: `wf_ac0cbb29-6bf`
- session_id: `fa4dc4eb-4705-47fd-b044-4f017b586b15`

Transcript

1. user (2026-06-19T15:52:06.172Z)

```
You are auditing a badminton rally-detection codebase for DOC/COMMENT STALENESS ? statements
that CONTRADICT the current code or data.
HARD RULES:
- Report ONLY factual staleness with CODE EVIDENCE (file:line or symbol) that proves it. Open
the cited code and confirm before reporting.
- NEVER invent a fact or a status. If a thing is built-but-default-OFF / unwired / unmeasured,
say EXACTLY that ? do not overclaim "in production".
- A "[design]" tag is stale ONLY if the thing is actually BUILT in code (cite the symbol).
- DO NOT propose subjective rewrites, style/tone changes, or improvements. DO NOT relitigate
owner-gated DECISIONS (owned-model roadmap, licensing/MIT-Apache stance, product/strategy
choices, genuinely-still-design proposals).
- Preserve the document's voice and formatting; make the SMALLEST surgical edit. stale_text MUST
be copied verbatim with enough context to be unique.
- IGNORE docs/archives/** (terminal snapshots) and any memory/ file.
Working dir is the repo root; use Read/Grep/Glob freely.

ADVERSARIALLY VERIFY these proposed staleness fixes for ONE file:
docs/RALLY_QUALITY_RESEARCH.md.
For EACH proposed fix: (1) open docs/RALLY_QUALITY_RESEARCH.md and confirm stale_text appears
VERBATIM (tighten/expand it to an exact, UNIQUE match if needed ? this text will be applied by
exact string match, so it must be perfect); (2) re-open the cited code yourself and confirm the
claim is genuinely contradicted by the CURRENT code; (3) confirm corrected_text is accurate,
preserves voice, and is minimal; (4) DROP it if it is subjective, an owner-gated decision,
already-correct, or you cannot verify the evidence. Output ONLY the survivors (verdict confirm
or revise), with stale_text guaranteed to be an exact unique substring of the file.

PROPOSED FIXES:
[
  {
    "file": "docs/RALLY_QUALITY_RESEARCH.md",
    "stale_text": "- **Corpus today: 6 golden videos** (label-set hash `50d98ffbc370`); ~10 more
expected via the\n  Roora vendoring track
([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)).",
    "corrected_text": "- **Corpus today: 15 golden videos** (started at 6, grown via the
multicourt/Boxhill clips); more\n  expected via the Roora vendoring track
([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)).",
    "why_stale": "Explicit present-tense claim 'Corpus today: 6 golden videos', but
output/human_lovo_manifest.json now has 15 entries. The §7.2 table rows measured at n=6 are
historical and left intact; this is the doc's only present-tense current-corpus-size
statement.",
    "severity": "medium"
  },
  {
    "file": "docs/RALLY_QUALITY_RESEARCH.md",
```

    "stale_text": "slow serve prep) ? those frames read as *active*. The 2 s merge then bridges the brief gaps, and\nbecause there is **no `max_window_duration` cap and no inactivity/boundary split**, a single\nwindow grows until the shuttle finally goes invisible ? which, at 96?99 % visibility, is rarely.\nResult: rally + dead-time + rally + ? collapse into a few mega-windows.",
    "corrected_text": "slow serve prep) ? those frames read as *active*. The 2 s merge then bridges the brief gaps, and\nbecause (at the time this was diagnosed) there was **no `max_window_duration` cap and no\ninactivity/boundary split**, a single window grew until the shuttle finally went invisible ? which,\nat 96?99 % visibility, is rarely. Result: rally + dead-time + rally + ? collapse into a few\nmega-windows. *(This is the failure W1/R4 fix; both are now default-ON since the R1 milestone, PR\n#204 ? `max_window_duration=6`, `window_inactivity_end=1.5` in `IndexingConfig`.)*",
    "why_stale": "Present-tense \"there is no `max_window_duration` cap and no inactivity/boundary split\" describes the SERVED operating point as pre-flip. Contradicted by backend/config/models.py:139 (`max_window_duration: float = 6.0`) and :153 (`window_inactivity_end: float = 1.5`), both flipped default-ON by the R1 milestone (docs/R1_MILESTONE_LAUNCH_REPORT.md, PR #204, LAUNCHED 2026-06-18). windowing.py::_served_preset_from_defaults() lifts these into the SERVED preset, so the served path now DOES cap+split.",
    "severity": "medium"
  }
]

2. assistant (2026-06-19T15:52:09.004Z)

I'll verify each proposed fix by checking the document and the cited code evidence.

3. user (2026-06-19T15:52:12.528Z)

fatal: Cannot fast-forward to multiple branches.

4. user (2026-06-19T15:52:12.669Z)

```
146-- `gt_loader.py` ? one canonical GT loader (both legacy formats). `fusion_golden.py` /
147-  `fusion_compare.py` / `fusion_audio.py` ? extract replayable per-window feature JSONs
148-  (shuttle 5 + person 5 + audio 3) so ablation/compare re-score on CPU without re-detecting.
149:- **Corpus today: 6 golden videos** (label-set hash `50d98ffbc370`); ~10 more expected via the
150-  Roora vendoring track ([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)).
151-
152-**Signals available to L2 (built, default-OFF):** net-crossing count (`rally_gate.py`, Gate-A),
```

5. user (2026-06-19T15:52:12.724Z)

```
85-- Windows shorter than `min_action_window_duration` (2.0 s) are dropped.
86-
87-**The conceptual flaw: "the shuttle is moving" is treated as "a rally is in progress."** During
88-dead time the shuttle is still visible and gently moving (being picked up, walked back, knock-ups,
89-slow serve prep) ? those frames read as *active*. The 2 s merge then bridges the brief gaps, and
90:because (at the time this was diagnosed) there was **no `max_window_duration` cap and no
91-inactivity/boundary split**, a single window grew until the shuttle finally went invisible ? which,
92-at 96?99 % visibility, is rarely. Result: rally + dead-time + rally + ? collapse into a few
93-mega-windows. *(This is the failure W1/R4 fix; both are now default-ON since the R1 milestone, PR
94:#204 ? `max_window_duration=6`, `window_inactivity_end=1.5` in `IndexingConfig`.)*
95-
96-> **Correction to an earlier hypothesis (honesty note).** A first pass guessed the cause was
97-> "velocity threshold not fps-invariant." It IS fps-invariant (the `fps/dframes` factor). The
98-> real cause is the *motion-equals-rally* conflation + 2 s bridging + no upper bound. fps
```

matters
99-> only secondarily: denser 60 fps sampling keeps more slow-motion dead-time frames above the
--
434- one-command rally-segmentation eval entrypoint (§4.5). **Success:** the current heuristic's
435- over-merge shows up as a *number* (low Edit, high merge-rate) on the 6 golden videos, and a
436- baseline row lands in the leaderboard. *(Low effort, high confidence ? research-backed §5.4.)*
437-
438-### Tier 1 ? Low-effort, high-confidence wins
439:- **W1 · Bounded windowing (the direct over-merge fix).** Add a `max_window_duration` cap and a
440- **sustained-inactivity split** to `trajectory_to_action_windows` ? port See et al.'s rule (split /
441- end a window when ?~80% of consecutive frames over a trailing window show no shuttle motion).
442- Config-gated, default preserves current behavior. **Success:** segmental Edit + F1@k improve vs
443- `CONTROL` on the golden set (?F1 CI excludes 0 **and** sign-test agrees) with **no served-windowing
444- regression**; merge-rate drops materially. *(Win ? §5.2 See et al., ~95% end-of-rally acc.)*
--
532-Status: ? not started · ? in progress · ? done. (Each row = its own default-OFF, measured PR.)
533-
534-| ID | Deliverable | Tier | Depends on | Acceptance / promote criterion | Status |
535-|---|---|---|---|---|---|
536-| **E1** | merge/split-rate metric + F1@k surfaced + one-command `rally_seg_eval` entrypoint; **froze baseline** (#185) | 0 | ? | over-merge shows up as a number on the 6 golden videos | ? baseline F1 0.300, merge_rate 0.523 |
537:| **W1** | Bounded windowing ? `max_window_duration` cap + inactivity split (See et al.) (#187) | 1 | E1 | seg F1@k ? vs baseline (CI?0 **and** sign-test), no served regression; merge-rate ? | ? **?F1 +0.139 (6/0), merge_rate 0.523?0.038** (default-OFF) |
538-| **W1.5** | Hard-cap fallback (`window_hard_cap`) ? chop continuously-active windows at the cap when no inactivity gap exists | 1 | W1 | golden ?F1 ?0 (or neutral) **+** fixes a real-footage over-merge it can't otherwise cap | ? **golden ?F1 +0.006 (n.s.)**; real `mahadevpura-1` 174s blob ? 24×?9s; retained **default-OFF** (real-footage + true-cap rationale, not a golden lift) |
539-| **W2** | Rally-state features (net-crossings ? #188 + **never-empty safeguard** ?; two-sided motion, **direction-reversal/peak**, audio = follow-on) | 1 | E1, W1 | ?1 cue ?F1 CI clears 0 + sign-test, survives served re-check ? promote; others retained default-OFF | ? **net-crossings ?F1 +0.019 (6/0)**, gate now **do-no-harm** (never zeroes a video); multi-feature scorer next (GPU) |
540-| **W3** | Keep scorer parameter-light (standing decision; TemporalMaxer evidence) | 1 | ? | documented; no learned temporal net until corpus ? now | ? |
541-| **M1** | Serve-start classifier ? rally-start boundary (pose/court) | 2 | W1, **court-mask decision (owner-gated)** | start-boundary timing error ?; promote on seg-F1 lift, no served regression | ? |
542-| **M2** | Per-venue threshold calibration vs golden | 2 | E1 | multi-court F1 ? toward Gen-0 0.561 without harming clean-court | ? |

6. assistant (2026-06-19T15:52:15.581Z)

Both stale_text blocks have already been partially edited. Let me check the current state carefully. The second fix's `stale_text` does NOT appear verbatim ? the file already contains the corrected version. Let me verify both proposed fixes against the actual file content.

7. user (2026-06-19T15:52:17.487Z)

82   exceeds `velocity_threshold` (default 0.02). Velocity = `(dist / frame_width) * (fps / dframes)`
83   ? **already fps-normalized** (? fraction of frame-width per second).
84   - Active frames within `dead_time_merge_gap` (2.0 s) are **merged** into one window.

```
85      - Windows shorter than `min_action_window_duration` (2.0 s) are dropped.
86
87      **The conceptual flaw: "the shuttle is moving" is treated as "a rally is in progress."** During
88      dead time the shuttle is still visible and gently moving (being picked up, walked back, knock-ups,
89      slow serve prep) ? those frames read as *active*. The 2 s merge then bridges the brief gaps, and
90      because (at the time this was diagnosed) there was **no `max_window_duration` cap and no
91      inactivity/boundary split**, a single window grew until the shuttle finally went invisible ? which,
92      at 96?99 % visibility, is rarely. Result: rally + dead-time + rally + ? collapse into a few
93      mega-windows. *(This is the failure W1/R4 fix; both are now default-ON since the R1 milestone, PR
94      #204 ? `max_window_duration=6`, `window_inactivity_end=1.5` in `IndexingConfig`.)*
95
96      > **Correction to an earlier hypothesis (honesty note).** A first pass guessed the cause was
97      > "velocity threshold not fps-invariant." It IS fps-invariant (the `fps/dframes` factor). The
98      > real cause is the *motion-equals-rally* conflation + 2 s bridging + no upper bound. fps matters
99      > only secondarily: denser 60 fps sampling keeps more slow-motion dead-time frames above the
100     > speed threshold. The `_LOCAL_vs_GEMINI_REPORT.md` has been corrected to match.
101
```

8. user (2026-06-19T15:52:18.042Z)

```
143       golden-set no-regression baseline with a GT-hash to detect label changes.
144
145     **Golden substrate**
146     - `gt_loader.py` ? one canonical GT loader (both legacy formats). `fusion_golden.py` /
147       `fusion_compare.py` / `fusion_audio.py` ? extract replayable per-window feature JSONs
148       (shuttle 5 + person 5 + audio 3) so ablation/compare re-score on CPU without re-detecting.
149     - **Corpus today: 6 golden videos** (label-set hash `50d98ffbc370`); ~10 more expected via the
150       Roora vendoring track ([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)).
151
152     **Signals available to L2 (built, default-OFF):** net-crossing count (`rally_gate.py`, Gate-A),
153     5 person/court features (`person_detector.py` + `fusion_features.py`), 3 audio-hit features
154     (`audio_features.py`). Fusion hooks: `fusion_hybrid.py::_enrich_candidate_stats` /
```

9. assistant (2026-06-19T15:52:22.415Z)

Key finding: **The second proposed fix is already applied to the file.** Lines 90-94 already contain the "corrected_text" version (past tense, with the R1 milestone note). The `stale_text` in proposal #2 (present-tense "there is **no `max_window_duration`...**") does NOT appear verbatim in the file. So proposal #2 must be DROPPED ? it's already correct.

Now let me verify proposal #1. The `stale_text` does appear. Let me verify the claim about the corpus size by checking the manifest cited in `why_stale`.

10. user (2026-06-19T15:52:24.640Z)

```
19june.db
19june_extract.log
19june_extract_rerun.log
19june_local_vs_gemini.md
19june_work
```

```
Badminton_BXH_2.rallies.csv
Badminton_BXH_2_golden_features.json
Boxhill_Doubles.rallies.csv
Boxhill_Doubles_golden_features.json
GX010125_highlights.mp4
GX010125_highlights_min5.mp4
GX010125_run
GX010128.rallies.csv
GX010128_golden_features.json
GX010137.rallies.csv
GX010137_golden_features.json
GX010141.rallies.csv
GX010141_golden_features.json
GX020094.rallies.csv
GX020094_golden_features.json
GX030094.rallies.csv
GX030094_golden_features.json
KushagraAviHighlight-Trailor.mp4
MahadevpuraSingles_highlights.mp4
REAL_FOOTAGE_VALIDATION_REPORT.md
_phase1_regress.log
_phase1_suite.log
_phase1_suite_fixed.log
_work19
ab_experiment_n6.json
adarsh_avi_full_vs_human.csv
adarsh_avi_singles_golden_features.json
adarsh_avi_vs_human.csv
adarsh_baseline_gated.csv
adarsh_baseline_segments.csv
adarsh_gemini_fullvideo.csv
adarsh_gemini_refined.csv
adarsh_gemini_smoke.csv
adarsh_min5s_segments.csv
adarsh_tuned_segments.csv
---MANIFEST---
output/human_lovo_manifest.json
```

11. user (2026-06-19T15:52:24.712Z)

```
Found 2 files
docs\RALLY_QUALITY_RESEARCH.md
eval_baselines\experiment_leaderboard.json
```

12. assistant (2026-06-19T15:52:26.656Z)

Let me verify the manifest content to confirm the corpus size claim.

13. user (2026-06-19T15:52:27.628Z)


```
[
  {
    "name": "adarsh_avi_singles",
    "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/adarsh_avi_full_traj.csv",
    "fps": 59.94,
    "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/Adarsh and Avi.rallies.csv"
  },
  {
    "name": "kushagra_singles",
    "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/kushagra_traj.csv",
```

```
      "fps": 59.94,
      "frame_width": 1920,
      "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/KushagraYashAviFirstVideo.rallies.csv"
    },
    {
      "name": "largetest_doubles",
      "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/largetestvideo_traj.csv",
      "fps": 29.97,
      "frame_width": 1920,
      "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/LargeTestVideo.rallies.csv"
    },
    {
      "name": "gbaaddy",
      "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/gbaaddy_traj.csv",
      "fps": 59.94,
      "frame_width": 1920,
      "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/gBaaddy20May2026.rallies.csv"
    },
    {
      "name": "testlarge_short",
      "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/testlarge_traj.csv",
      "fps": 29.97,
      "frame_width": 1920,
      "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/TestLargeVideo.rallies.csv"
    },
    {
      "name": "mahadevpura_singles",
      "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/mahadevpura_traj.csv",
      "fps": 29.97,
      "frame_width": 1920,
      "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/MahadevpuraSingles.rallies.csv"
    },
    {
      "name": "mahadevpura_2",
      "fps": 59.94,
      "frame_width": 1920.0,
      "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/mahadevpura_2.rallies.csv",
      "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/mahadevpura2_june12_traj.csv"
    },
    {
      "name": "GX010128",
      "fps": 59.94,
      "frame_width": 1920.0,
      "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/GX010128.rallies.csv",
      "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/GX010128_june12_traj.csv"
    },
    {
      "name": "mahadevpura_1",
      "fps": 59.94,
      "frame_width": 1920.0,
      "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/mahadevpura_1.rallies.csv",
```

```
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/mahadevpura1_june12_traj.csv"
      },
      {
        "name": "GX030094",
        "fps": 29.97,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/GX030094.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/GX030094_june15_traj.csv"
      },
      {
        "name": "Badminton_BXH_2",
        "fps": 59.94,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/Badminton_BXH_2.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/Badminton_BXH_2_june15_traj.csv"
      },
      {
        "name": "GX010141",
        "fps": 59.94,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/GX010141.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/GX010141_june17_traj.csv"
      },
      {
        "name": "GX010137",
        "fps": 59.94,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/GX010137.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/GX010137_june17_traj.csv"
      },
      {
        "name": "Boxhill_Doubles",
        "fps": 29.97,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/Boxhill_Doubles.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/Boxhill_Doubles_june18_traj.csv"
      },
      {
        "name": "GX020094",
        "fps": 29.97,
        "frame_width": 1920.0,
        "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/GX020094.rallies.csv",
        "trajectory": "C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/GX020094_june18_traj.csv"
      }
    ]
```


14. user (2026-06-19T15:52:28.913Z)

```
db96ddc docs: capture external due-diligence review inputs (research, pick-up-later) (#207)
bb4d6e2 feat(eval): Tier 1 (W1.5) ? hard-cap fallback enforces the window cap on continuously-active trajectories (#195)
950106b fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-empty video (#188
```

follow-up) (#190)
a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0) (#187)

15. assistant (2026-06-19T15:52:36.738Z)

The manifest has 15 entries. Now I need to verify the corpus-size claim more carefully. The proposal claims the doc's "Corpus today: 6 golden videos" is stale because the manifest has 15. But I must be careful: the doc text also includes a specific label-set hash `50d98ffbc370`, and the §7.2 measurements are explicitly at n=6.

Let me check whether the corpus that this doc refers to (the golden eval substrate, keyed by that hash) is genuinely 15 now, or whether the manifest is a different/superset thing. Let me check what generates that hash and whether the eval actually runs on 15 videos now.

16. user (2026-06-19T15:52:37.678Z)

```
Found 16 files
tests\test_ablation.py
tests\test_experiment.py
backend\eval\rally_seg_eval.py
backend\eval\fusion_audio.py
backend\eval\experiment.py
backend\eval\ablation.py
tests\test_served_gate_a_regression.py
backend\eval\ablation_pdf.py
scripts\nightly_regression.py
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\lovo_resweep.py
training\gen0\harness.py
backend\eval\serve_contrast.py
backend\eval\served_gate_a.py
backend\eval\fusion_golden.py
tests\test_experiment_honest.py
backend\eval\rally_seq_proto.py
```

17. user (2026-06-19T15:52:39.008Z)

```
35:| Path | Mahadevpura (clean, single court) | Kushagra (multi-court) |
43:multi-court / busy venues. The owned Gen-0 head already closes most of the multi-court gap
409:corpus growth to n?16 and **multi-court isolated as its own reported slice** (§6 Tier-3 /
§7) ? do
424:Gemini reference (today **0.93 single-court / 0.66 multi-court** tIoU-F1) ? but tracked
primarily
464:  it scored 0.157 on multi-court). Tune `velocity_threshold` + `merge_gap` so local segment
counts
465:  track golden counts per venue. **Success:** multi-court F1 rises toward the Gen-0 head's
0.561
508:- **Headline:** multi-court rally-segmentation **tIoU-F1**, `baseline ? final`, plus **gap-
closure %**
511:  heuristic?Gemini multi-court gap** ? i.e. local multi-court F1 ? ~0.41, ideally beating
Gen-0's 0.561.
542:| **M2** | Per-venue threshold calibration vs golden | 2 | E1 | multi-court F1 ? toward
Gen-0 0.561 without harming clean-court | ? |
```

18. assistant (2026-06-19T15:52:42.784Z)

There's an archive directory `2026-06-18-lovo-resweep-n13` indicating the corpus grew to at least n=13. Let me check the current corpus size used by the eval and what the doc itself says about corpus growth. Let me look at how the manifest is consumed and confirm the present operating corpus size. Let me also check line 409 context ("corpus growth to n?16").

19. user (2026-06-19T15:52:44.675Z)

```
402       upstream proposal distribution (the Gate-A episode) ? a citable framing and candidate paper "lesson."
403     - **Citable Layer-1 anchor (public benchmark, NOT our footage).** WASB SBDT F1 95.6 (step=1) / 94.0
404       (step=3) vs MonoTrack 92.1, TrackNetV2 89.4, BallSeg 71.7 (WASB, BMVC 2023). Justifies the frozen
405       detector choice; **keep distinct from our own, still-unmeasured footage domain-gap** (§2.1, a real
406       open limitation the review also flags).
407
408     **Sequencing:** these are *publication-prep + post-current-idea* levers. The prerequisites remain
409     corpus growth to n?16 and **multi-court isolated as its own reported slice** (§6 Tier-3 / §7) ? do
410     those first; the items above land once current windowing/fusion ideas are exhausted.
411
412     ---
413
414     ## 6. Prioritized roadmap (cheap ? durable)
415
416     Every item is **its own measured PR**: default-OFF, scored on the §4 metric set over the golden set
417     with honest stats (CI + sign-test), **re-confirmed on the served windowing** (the eval?serve gate),
418     logged in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md). Ordered by the research's
419     effort/confidence tiering. Each carries a **falsifiable success criterion** ? if it isn't met, the
420     item does not promote (and that null is itself recorded).
421
422     ### Definition of "bridging the gap"
423     Local rally segmentation, measured **honestly LOVO-by-match on the golden set**, approaches the
424     Gemini reference (today **0.93 single-court / 0.66 multi-court** tIoU-F1) ? but tracked primarily
425     through the **over-segmentation-sensitive metrics** (segmental Edit + F1@k + low merge/split rate),
426     since those, not bare tIoU-F1, capture our actual failure. A secondary, free signal is rising
427     **agreement-with-Gemini** on the unlabeled pool (§4.4, shadow only).
428
429     ### Tier 0 ? Make the failure measurable FIRST (do before any fix)
```

20. user (2026-06-19T15:52:45.962Z)

```
10-
11-Run (on the GPU box):
12:    python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir output
13-
14-Each output `output/<name>_golden_features.json` is the replayable feature substrate ? re-run
--
38-
39-def _derive_video_path(spec: Dict[str, Any]) -> str:
40:    """Original MP4 for a manifest entry ? explicit `video_path`, else derived from the
41-    golden labels path (``X.rallies.csv`` -> ``X.MP4``, the golden-set naming convention)."""
42-    if spec.get("video_path"):
--
127-
128-
129:def run(manifest_path: str, out_dir: str, frame_skip: int = 3, iou: float = 0.5,
130-        fresh: bool = False, smallest_first: bool = False) -> List[str]:
131:    with open(manifest_path, encoding="utf-8") as f:
132-        specs = json.load(f)
133-    if smallest_first:
--
```

```
180-def main() -> None:
181-    ap = argparse.ArgumentParser(description="Extract golden-set shuttle+person features
(fusion P3 GPU step).")
182:    ap.add_argument("--manifest", default="output/human_lovo_manifest.json")
183-    ap.add_argument("--out-dir", default="output")
184-    ap.add_argument("--frame-skip", type=int, default=3)
--
188-                    help="process smaller videos first for quick feedback (LOVO is order-
independent)")
189-    args = ap.parse_args()
190:    run(args.manifest, args.out_dir, args.frame_skip, args.iou, fresh=args.fresh,
191-        smallest_first=args.smallest_first)
192-
```

21. assistant (2026-06-19T15:52:51.594Z)

The eval substrate manifest (`output/human_lovo_manifest.json`, the default consumed by `fusion_golden.py`) now has 15 entries. The doc says line 409 "corpus growth to n?16" is still a future prereq, which is consistent with the corpus currently being below 16 ? so 15 fits well.

Now I have an important nuance to verify: the doc's line 149 ties "6 golden videos" to a specific label-set hash `50d98ffbc370`. Let me confirm whether that hash is still the current/meaningful hash, and check what the experiment_leaderboard says, since overclaiming the hash is wrong. The proposed correction drops the hash entirely. Let me verify the hash is now stale too.

22. user (2026-06-19T15:52:53.153Z)

```
3-    "timestamp": "2026-06-14T06:23:51.507393+00:00",
4-    "iou": 0.5,
5:    "label_set_hash": "50d98ffbc370",
6-    "num_videos": 6,
7-    "variant_a": {
--
49-    "timestamp": "2026-06-14T06:24:56.487711+00:00",
50-    "iou": 0.5,
51:    "label_set_hash": "50d98ffbc370",
52-    "num_videos": 6,
53-    "variant_a": {
--
95-    "timestamp": "2026-06-14T00:00:00+00:00",
96-    "iou": 0.5,
97:    "label_set_hash": "50d98ffbc370",
98-    "num_videos": 6,
99-    "promotion": {
```

23. user (2026-06-19T15:52:54.155Z)

EISDIR: illegal operation on a directory, read 'C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13'

24. assistant (2026-06-19T15:53:01.398Z)

The leaderboard's most recent entries (which feed §7.2 measurements) are still at `label_set_hash: 50d98ffbc370`, num_videos: 6 ? those are historical n=6 measurements. But the eval substrate (manifest) is now 15 videos, and the archive `2026-06-18-lovo-resweep-n13` confirms the corpus was already grown past 6 (to n=13, now 15).

Now there's a critical nuance about the proposed correction. The proposed `corrected_text` removes the label-set hash `50d98ffbc370`. But that hash is genuinely the hash of the 6-video label set, and it's still referenced live in the leaderboard and §7.2 measurements. The cleanest minimal fix should update the count while remaining honest. However, the proposed correction's number "15" and rationale are accurate. Let me check exactly what §7.2 references, to ensure removing the hash from line 149 doesn't break a cross-reference.

25. user (2026-06-19T15:53:03.132Z)

149:- **Corpus today: 6 golden videos** (label-set hash `50d98ffbc370`); ~10 more expected via the

26. user (2026-06-19T15:53:03.738Z)

```
130       "honest" stats block), `compare_unlabeled()` (agreement on unlabeled footage), `CONTROL`
131       (pinned baseline), `record_experiment()` ? `eval_baselines/experiment_leaderboard.json`.
132     - `ablation.py` ? leave-one-signal-out marginal ?F1 with CI + sign-test + a no-over-claim verdict
133       vocabulary (`earns_keep` / `suggestive` / `inert` / `structural_protected`).
134     - `classifier.py` ? dependency-free numpy `LogisticRegression` (the learned scorer; JSON-serializable).
135
136     **Eval?serve discipline + gates**
137     - `windowing.py` ? single-sources the **SERVED** preset from `IndexingConfig` defaults so eval
138       and production can't silently diverge; `PROPOSAL` (recall-oriented) is where the golden feature
139       substrate is built. (This exists *because* Gate-A measured **+0.074 on PROPOSAL but ?0.008 on
140       SERVED** and was reverted ? the canonical eval?serve lesson.)
141     - `promotion.py` / `served_gate_a.py` ? promote a cue only if it does NOT regress at the served
142       windowing; `per_user_gate.py` / `per_user_regression.py` ? per-user analogues; `regression.py` ?
143       golden-set no-regression baseline with a GT-hash to detect label changes.
144
145     **Golden substrate**
146     - `gt_loader.py` ? one canonical GT loader (both legacy formats). `fusion_golden.py` /
147       `fusion_compare.py` / `fusion_audio.py` ? extract replayable per-window feature JSONs
148       (shuttle 5 + person 5 + audio 3) so ablation/compare re-score on CPU without re-detecting.
149     - **Corpus today: 6 golden videos** (label-set hash `50d98ffbc370`); ~10 more expected via the
150       Roora vendoring track ([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)).
151
```

27. assistant (2026-06-19T15:53:11.487Z)

The hash `50d98ffbc370` appears only once in the doc (line 149), so removing it doesn't break any cross-reference. The line is a present-tense statement about the *current* corpus substrate, and that substrate (the default manifest consumed by the eval) is now 15 videos, not 6. This is genuinely stale.

Now let me assess the proposed `corrected_text` for #1. It says "started at 6, grown via the multicourt/Boxhill clips". The manifest confirms Boxhill clips (Boxhill_Doubles, Badminton_BXH_2) and various GX clips. The framing "multicourt/Boxhill clips" is reasonable and grounded. However, I'll refine to keep it tight and verifiable. The "~10 more expected" ? "more expected" is fine since the original "~10 more" was tied to reaching the old target.

Let me confirm the n=15 count once more and verify there isn't a more authoritative "served Gate-A" present-tense corpus reference, then finalize.

28. user (2026-06-19T15:53:13.070Z)

entries: 15
['adarsh_avi_singles', 'kushagra_singles', 'largetest_doubles', 'gbaaddy', 'testlarge_short', 'mahadevpura_singles', 'mahadevpura_2', 'GX010128', 'mahadevpura_1', 'GX030094', 'Badminton_BXH_2', 'GX010141', 'GX010137', 'Boxhill_Doubles', 'GX020094']

29. assistant (2026-06-19T15:53:21.159Z)

Confirmed: 15 entries in the default eval manifest. The "Corpus today: 6 golden videos" statement is genuinely stale.

Now let me finalize my assessment of both proposals:

Proposal #1 (corpus size): CONFIRMED stale. The `stale_text` appears verbatim at line 149. The eval substrate manifest (default consumed by `fusion_golden.py`) now has 15 entries. The doc itself (line 409) lists "corpus growth to n?16" as a still-pending prereq, consistent with 15 < 16. I'll **revise** the corrected_text slightly: the proposal's phrasing is good, but I'll keep it minimal and accurate. The proposal's wording "started at 6, grown via the multicourt/Boxhill clips" is grounded in the manifest. I'll accept it largely as-is.

Proposal #2 (windowing cap): DROP. The `stale_text` (present-tense "there is **no `max_window_duration` cap...**") does NOT appear verbatim in the file ? lines 90-94 already contain the corrected past-tense version with the R1 milestone note. The fix is already applied. There is nothing to change.

Let me double-check proposal #2's exact stale_text really is absent by searching for its distinctive opening fragment.

30. user (2026-06-19T15:53:23.528Z)

```
90:because (at the time this was diagnosed) there was **no `max_window_duration` cap and no
---check present-tense 'there is no'---
exit-grep-done
```

31. assistant (2026-06-19T15:53:28.554Z)

Confirmed. The present-tense phrasing "because there is **no `max_window_duration` cap and no**" from proposal #2's `stale_text` does NOT exist. The file already has the past-tense corrected version ("because (at the time this was diagnosed) there was **no...**" plus the R1 milestone note). Proposal #2 is already applied ? DROP it.

Final survivors: only proposal #1, with a minor revision to keep the corrected_text minimal and grounded.

32. user (2026-06-19T15:53:39.793Z)

Structured output provided successfully

33. assistant (2026-06-19T15:53:47.226Z)

I verified both proposed fixes for `docs/RALLY_QUALITY_RESEARCH.md`. One survives (revised), one dropped.

Survivor ? Corpus size (line 149), verdict: revise. "Corpus today: 6 golden videos" is genuinely stale. The live eval substrate `output/human_lovo_manifest.json` ? the default `--manifest` consumed by `backend/eval/fusion_golden.py` (`main()`, line 182) ? now lists **15** entries. The archive dir `docs/archives/quality-iterations/2026-06-18-lovo-resweep-n13` confirms it already grew past 6, and the doc's own line 409 still cites "corpus growth to n?16" as a pending prereq (consistent with 15). I revised the proposed correction to **keep** the label-set hash `50d98ffbc370` as the historical origin point ? it is the only occurrence in the doc and still keys the live n=6 numbers in `eval_baselines/experiment_leaderboard.json` and the §7.2 rows, so deleting it would drop a real anchor.

Dropped ? Windowing cap (lines 90-94). Already-correct. The proposed `stale_text` ("because there is **no `max_window_duration` cap and no**...", present tense) does not exist in the file. Lines 90-94 already contain the past-tense corrected version with the R1-milestone note (`max_window_duration=6`, `window_inactivity_end=1.5`, PR #204). Nothing to change.